

LAB EXPERIMENT RECORD

Set 9: Online Learning Activity Analysis

Department	B.Tech AIDS
Course Code / Course Name	DSA06 / Data Handling and Visualization
Experiment	Data Visualization using R
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Day	Saturday

Aim

To import and analyze online learning activity data in R using a histogram, boxplot, bubble scatter plot, and a monthly trend line with moving average smoothing, and to design an interactive mini dashboard using R Shiny.

Dataset: Online_Learning_Activity.csv

Student ID	Gender	Age	Course	Study Time (hrs)	Videos Watched	Quiz Score	Login Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

Note: The R programs first create Online_Learning_Activity.csv from the table above and then import it using read.csv().

PROGRAM 1: Histogram and Boxplot of Quiz Scores

Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.

```
# Create the Online_Learning_Activity.csv file
learning <- data.frame(
  Student_ID = c("L01", "L02", "L03", "L04", "L05", "L06"),
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),
  Age = c(20, 22, 19, 21, 23, 20),
  Course = c("R", "R", "SQL", "R", "R", "SQL"),
  Study_Time = c(3.5, 4.2, 2.0, 5.0, 2.5, 4.0),
  Videos_Watched = c(12, 15, 8, 18, 9, 14),
  Quiz_Score = c(78, 85, 65, 92, 70, 88),
  Login_Date = c("2025-01-05", "2025-01-05", "2025-02-08",
    "2025-02-08", "2025-03-12", "2025-03-12")
)
```

```

)
write.csv(learning, "Online_Learning_Activity.csv", row.names = FALSE)

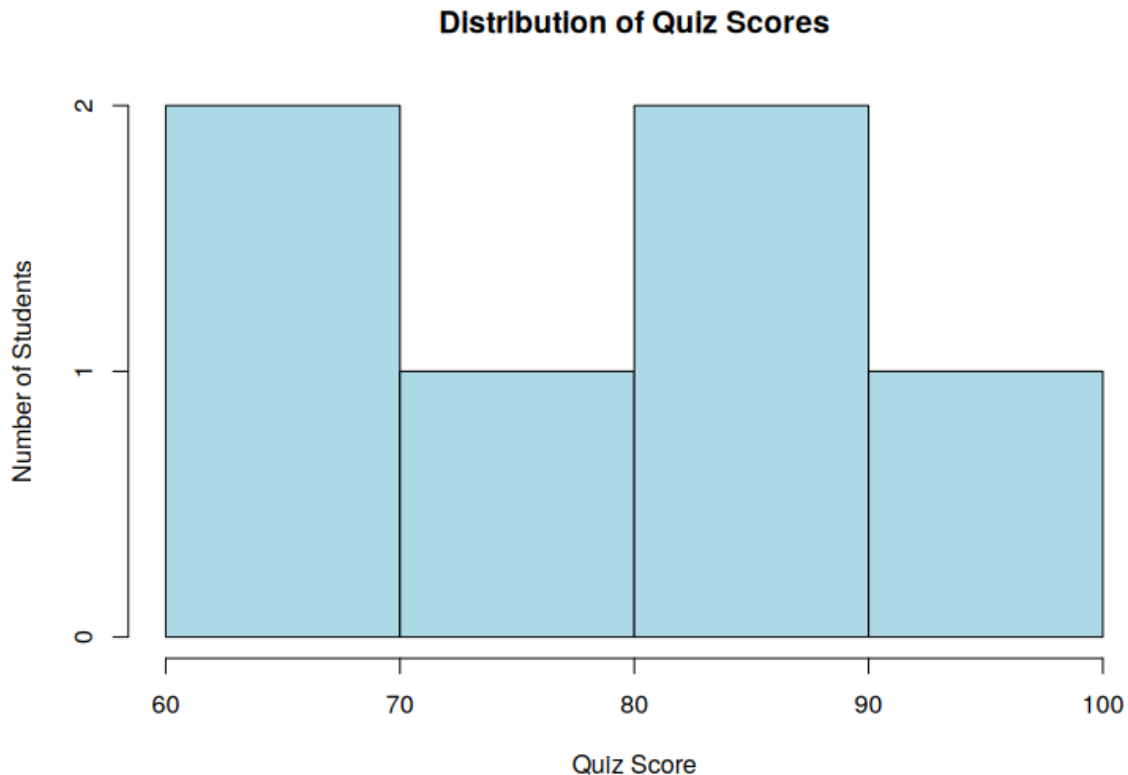
# Import the dataset
learning_data <- read.csv("Online_Learning_Activity.csv")
print(learning_data)

# 1(a). Histogram of Quiz Scores
hist(learning_data$Quiz_Score,
     breaks = seq(60, 100, by = 10),
     col = "lightblue",
     border = "black",
     xlab = "Quiz Score",
     ylab = "Number of Students",
     main = "Distribution of Quiz Scores")

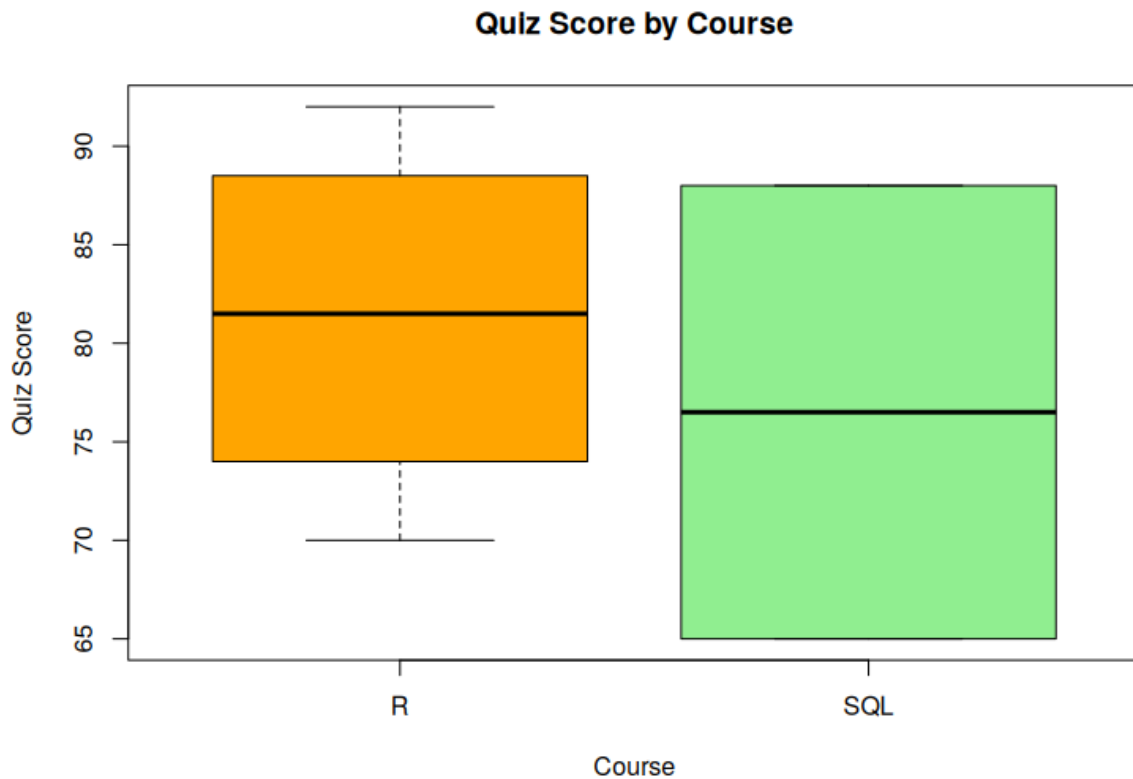
# 1(b). Boxplot of Quiz Score by Course
boxplot(Quiz_Score ~ Course,
       data = learning_data,
       col = c("orange", "lightgreen"),
       xlab = "Course",
       ylab = "Quiz Score",
       main = "Quiz Score by Course")

# Average score per course
print(aggregate(Quiz_Score ~ Course, data = learning_data, FUN = mean))

```



Output 1(a): Distribution of Quiz Scores



Output 1(b): Quiz Score by Course

Interpretation: Quiz scores range from 65 to 92 with an average of 79.7. Two students scored 60–70, one scored 70–80, two scored 80–90, and one scored above 90. In the boxplot, R students have a higher median score (81.5) and average (81.25) than SQL students (median and average 76.5), but the SQL scores are spread much more widely, from 65 to 88. Overall, most students performed moderately well, and R students were slightly more consistent.

PROGRAM 2: Bubble Scatter Plot of Study Time vs Quiz Score

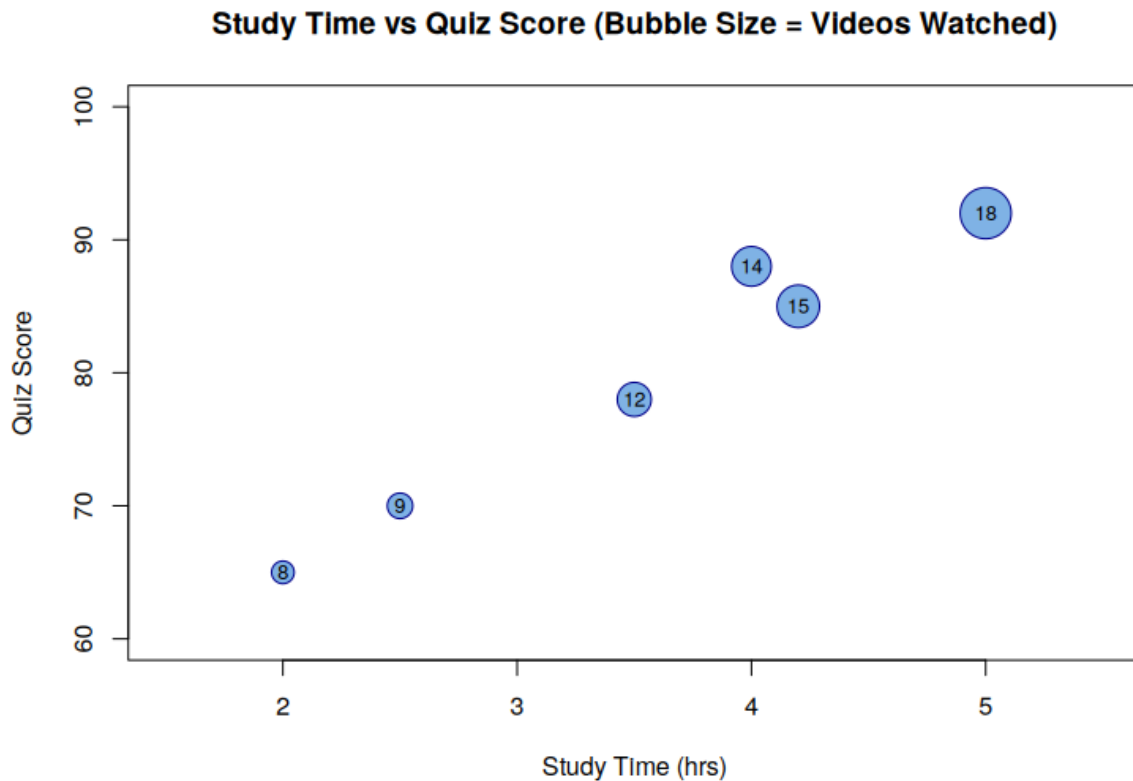
Create a scatter plot using R programming between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.

```
# Create the Online_Learning_Activity.csv file
learning <- data.frame(
  Student_ID = c("L01", "L02", "L03", "L04", "L05", "L06"),
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),
  Age = c(20, 22, 19, 21, 23, 20),
  Course = c("R", "R", "SQL", "R", "R", "SQL"),
  Study_Time = c(3.5, 4.2, 2.0, 5.0, 2.5, 4.0),
  Videos_Watched = c(12, 15, 8, 18, 9, 14),
  Quiz_Score = c(78, 85, 65, 92, 70, 88),
  Login_Date = c("2025-01-05", "2025-01-05", "2025-02-08",
    "2025-02-08", "2025-03-12", "2025-03-12")
)
write.csv(learning, "Online_Learning_Activity.csv", row.names = FALSE)

# Import the dataset
learning_data <- read.csv("Online_Learning_Activity.csv")
print(learning_data)

# 2. Bubble Scatter Plot: Study Time vs Quiz Score
plot(learning_data$Study_Time, learning_data$Quiz_Score,
  pch = 21,
  bg = rgb(0, 0.4, 0.8, alpha = 0.5), # transparency
  col = "darkblue",
  cex = learning_data$Videos_Watched / 4, # bubble size
  xlim = c(1.5, 5.5),
  ylim = c(60, 100),
  xlab = "Study Time (hrs)",
  ylab = "Quiz Score",
  main = "Study Time vs Quiz Score (Bubble Size = Videos Watched)")
text(learning_data$Study_Time, learning_data$Quiz_Score,
  labels = learning_data$Videos_Watched, cex = 0.8)

# Correlation between study time and quiz score
cor(learning_data$Study_Time, learning_data$Quiz_Score)
```



Output 2: Study Time vs Quiz Score (Bubble Size = Videos Watched)

Learning Pattern: The bubbles rise from the bottom left to the top right, showing a strong positive relationship between study time and quiz score (correlation about 0.98). The bubbles also get larger as scores increase, so students who watched more videos scored higher (correlation about 0.97). The student who studied 5 hours and watched 18 videos scored the highest (92), while the student who studied 2 hours and watched 8 videos scored the lowest (65).

PROGRAM 3: Monthly Trend with Moving Average

Using R programming, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.

```
# Create the Online_Learning_Activity.csv file
learning <- data.frame(
  Student_ID = c("L01", "L02", "L03", "L04", "L05", "L06"),
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),
  Age = c(20, 22, 19, 21, 23, 20),
  Course = c("R", "R", "SQL", "R", "R", "SQL"),
  Study_Time = c(3.5, 4.2, 2.0, 5.0, 2.5, 4.0),
  Videos_Watched = c(12, 15, 8, 18, 9, 14),
  Quiz_Score = c(78, 85, 65, 92, 70, 88),
  Login_Date = c("2025-01-05", "2025-01-05", "2025-02-08",
                 "2025-02-08", "2025-03-12", "2025-03-12")
)
write.csv(learning, "Online_Learning_Activity.csv", row.names = FALSE)

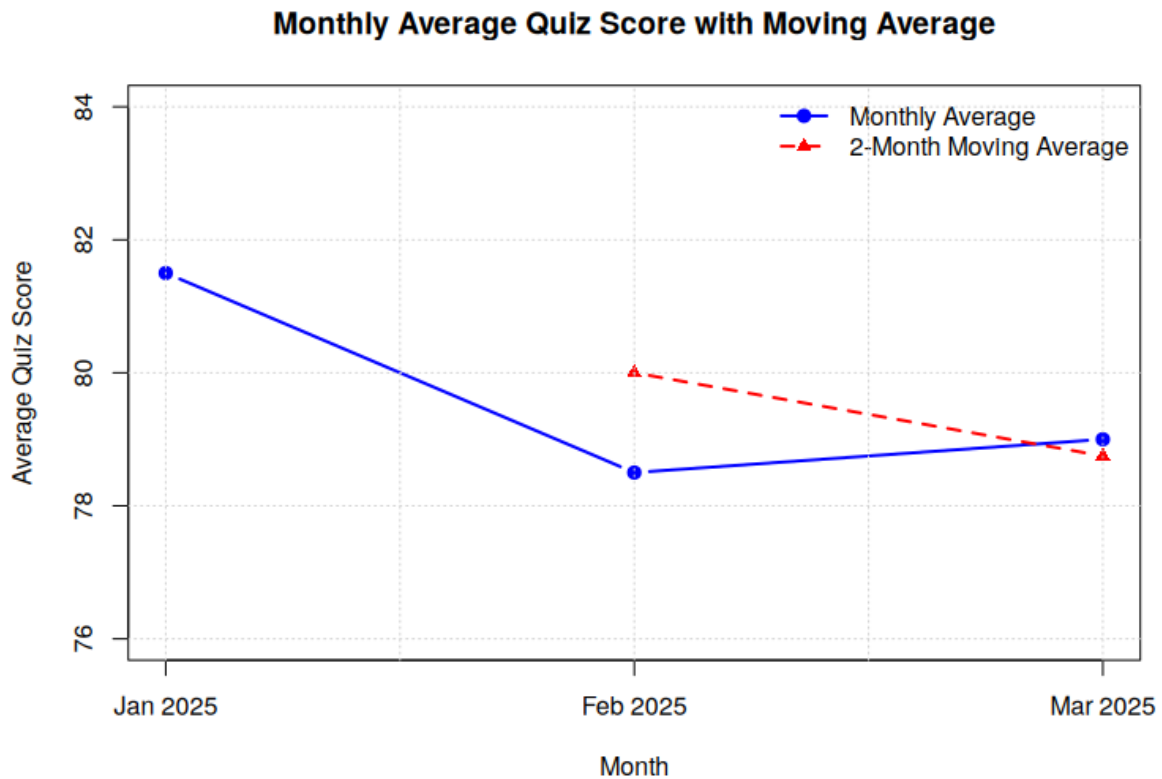
# Import the dataset
learning_data <- read.csv("Online_Learning_Activity.csv")
print(learning_data)

# Convert Login_Date into Date format
learning_data$Login_Date <- as.Date(learning_data$Login_Date)
learning_data$Month <- format(learning_data$Login_Date, "%Y-%m")

# Average quiz score per month
monthly_avg <- aggregate(Quiz_Score ~ Month, data = learning_data, FUN = mean)

# 2-month moving average
monthly_avg$Moving_Avg <- as.numeric(stats::filter(monthly_avg$Quiz_Score,
                                                    rep(1/2, 2), sides = 1))
print(monthly_avg)

# 3. Line Chart with Moving Average Smoothing
month_no <- 1:nrow(monthly_avg)
plot(month_no, monthly_avg$Quiz_Score,
     type = "b", col = "blue", lwd = 2, pch = 19,
     xaxt = "n", ylim = c(76, 84),
     xlab = "Month", ylab = "Average Quiz Score",
     main = "Monthly Average Quiz Score with Moving Average")
axis(1, at = month_no, labels = c("Jan 2025", "Feb 2025", "Mar 2025"))
lines(month_no, monthly_avg$Moving_Avg, type = "b",
     col = "red", lwd = 2, lty = 2, pch = 17)
legend("topright", legend = c("Monthly Average", "2-Month Moving Average"),
     col = c("blue", "red"), lty = c(1, 2), pch = c(19, 17), lwd = 2, bty = "n")
grid()
```



Output 3: Monthly Average Quiz Score with Moving Average

Trend: The average quiz score is 81.5 in January, drops to 78.5 in February, and rises slightly to 79.0 in March. Since there are only three months of data, a 2-month moving average was used, giving 80.0 for February and 78.75 for March. The smoothed line shows a slight decreasing trend in average scores over the three months.

PROGRAM 4: Interactive Mini Dashboard for Online Learning Activity

Import Online_Learning_Activity.csv into Tableau. Create a bar chart showing average quiz score by Course. Create a scatter plot (Study Time vs Quiz Score). Add filter for Gender and Course. Design a mini interactive dashboard.

Note: Tableau could not be installed on the system used for this experiment, so the interactive dashboard was developed in R using the Shiny package, with the same charts and interactive filters required in the question.

```
library(shiny)

# Dataset for Set 9 (Online_Learning_Activity.csv)
learning_data <- data.frame(
  Student_ID = c("L01", "L02", "L03", "L04", "L05", "L06"),
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),
  Course = c("R", "R", "SQL", "R", "R", "SQL"),
  Study_Time = c(3.5, 4.2, 2.0, 5.0, 2.5, 4.0),
  Quiz_Score = c(78, 85, 65, 92, 70, 88)
)
course_cols <- c(R = "orange", SQL = "seagreen")

ui <- fluidPage(
  titlePanel("Online Learning Activity Dashboard"),
  sidebarLayout(
    sidebarPanel(width = 3,
      checkboxGroupInput("gender", "Gender:", choices = c("Female", "Male"),
        selected = c("Female", "Male")),
      checkboxGroupInput("course", "Course:", choices = c("R", "SQL"),
        selected = c("R", "SQL"))
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("barPlot", height = "340px")),
        column(6, plotOutput("scatterPlot", height = "340px"))
      )
    )
  )
)

server <- function(input, output) {
  filtered <- reactive({
    subset(learning_data, Gender %in% input$gender & Course %in% input$course)
  })

  output$barPlot <- renderPlot({
    avg <- aggregate(Quiz_Score ~ Course, data = filtered(), FUN = mean)
    bp <- barplot(avg$Quiz_Score, names.arg = avg$Course, col = course_cols[avg$Course],
      ylim = c(0, 100), xlab = "Course", ylab = "Average Quiz Score",
      main = "Average Quiz Score by Course")
    text(bp, avg$Quiz_Score, labels = round(avg$Quiz_Score, 2), pos = 3)
  }, width = 420, height = 340, res = 96)

  output$scatterPlot <- renderPlot({
    d <- filtered()
    plot(d$Study_Time, d$Quiz_Score, pch = 19, cex = 1.6,
      col = adjustcolor(course_cols[d$Course], alpha.f = 0.8),
      xlim = c(1.5, 5.5), ylim = c(60, 100),
```

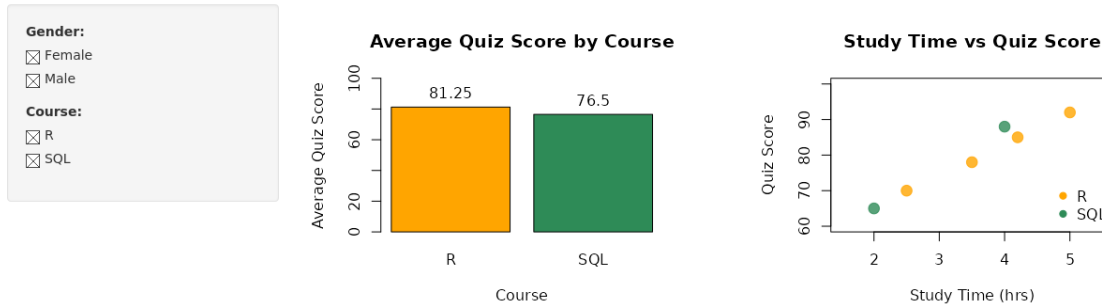
```

      xlab = "Study Time (hrs)", ylab = "Quiz Score", main = "Study Time vs Quiz Score")
      legend("bottomright", legend = names(course_cols), col = course_cols, pch = 19, bty = "n")
    }, width = 420, height = 340, res = 96)
  }

shinyApp(ui = ui, server = server)

```

Online Learning Activity Dashboard



Output 4: Online Learning Activity Interactive Dashboard (R Shiny)

Observation: The mini dashboard shows the average quiz score by course (R: 81.25, SQL: 76.5) and the study time vs quiz score scatter plot together. The Gender and Course filters update both charts, so performance can be compared for male and female students and for each course.

Procedure

1. Create the dataset, save it as `Online_Learning_Activity.csv`, and import it using `read.csv()`.
2. Use `hist()` to plot the quiz score distribution and `boxplot()` to compare quiz scores by course.
3. Use `plot()` with `cex` based on `Videos_Watched` and a transparent `rgb()` colour to draw the bubble scatter plot.
4. Convert `Login_Date` using `as.Date()`, extract the month with `format()`, and calculate the monthly average using `aggregate()`.
5. Apply a 2-month moving average using `stats::filter()` and plot it with `lines()` on the monthly line chart.
6. Install and load the shiny package, and create the dashboard layout using `fluidPage()`, sidebar filters, and output areas for the charts.
7. Write the server function to filter the data with `reactive()` and redraw the outputs, then run the dashboard using `shinyApp()` and test the filters.

Result

The online learning activity data was successfully imported and visualized in R using a histogram, boxplot, bubble scatter plot, and a monthly trend line with moving average smoothing, and an interactive mini dashboard was designed using R Shiny.

Key R Functions Used

Function	Purpose
<code>write.csv()</code> / <code>read.csv()</code>	Creates and imports the CSV dataset.
<code>hist()</code>	Creates the quiz score histogram.
<code>boxplot()</code>	Compares quiz scores by course.
<code>rgb(alpha =)</code>	Applies transparency to the bubbles.
<code>as.Date()</code> / <code>format()</code>	Converts dates and extracts the month.
<code>aggregate()</code>	Calculates average quiz score per month and per course.
<code>stats::filter()</code>	Calculates the moving average.
<code>lines()</code> / <code>legend()</code>	Adds the moving average line and legend.
<code>fluidPage()</code> / <code>checkboxGroupInput()</code>	Creates the dashboard layout and filters.
<code>reactive()</code> / <code>renderPlot()</code>	Filters the data and redraws the charts.
<code>shinyApp()</code>	Runs the interactive dashboard.